Name ____ Date ___________ Period _____

Modeling with a Parent Function: the Open Box

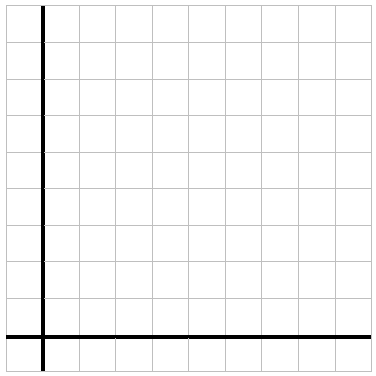
Algebra II | Unit 1 | Day 9 | TEKS A2.2.A, A2.7.I

Objective: I can build a function that models a real situation, decide which domain makes sense for the context, and use a table or graph to find a maximum.

The Problem

Start with a flat sheet of cardboard **20 inches by 10 inches**. Cut a square of side **x** out of each of the four corners, then fold the flaps up to make an open-top box.

Sketch the sheet with the corner squares marked, and label one side x.



Part 1 — Build the Function

Each cut removes x from BOTH ends of a side, so each side loses 2x.

Height of the box: ______________________

Length of the box: ______________________ Width of the box: ______________________

Volume V(x) = length · width · height = _________________________________

Multiplied out: V(x) = _________________________________ Which parent function family? _______________

Part 2 — Find the Domain That Makes Sense

The equation accepts any real number. The cardboard does not.

Why must x be greater than 0? ____

The short side is 10 inches. What happens if x = 5? ____

So the contextual domain is ______________________ interval _________________ Open or closed? __________________________

Part 3 — Make a Table

Compute $V(x) = x(20 - 2x)(10 - 2x)$ for each value. Round to the nearest tenth.

x	0.5	1	1.5	2	2.5	3	4	4.5
V(x)								

Between which two x-values does the volume stop growing and start shrinking? __________________________

Largest volume in your table: __________________________ Is that certainly the true maximum? ____________________________

Part 4 — Refine the Maximum

Test values between 2 and 2.5 to close in.

x	2.0	2.1	2.11	2.2	2.3
V(x)					

Maximum volume ~ ____________________ when x ~ ________________

Box dimensions then: ______________________________

Range of the function on this domain: __________________________ interval __________________

Part 5 — Interpret in Context

1. What does V(1) = 144 mean in words? __________________________________
2. Why is x = 5 excluded but x = 4.5 allowed? __________________________________
3. Would a domain of all reals make sense here? __________________________________

Practice — A Second Box

A sheet is 16 in. by 8 in. Squares of side x are cut from the corners. Show your work.

1. Write the volume function.

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V(x) = ______________________________

2. Which side limits the domain, and what is the limit? ____________________________

3. Contextual domain: __________________________

4. Compute V(1) and V(2). Show your arithmetic.

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V(1) = __________________ V(2) = __________________

5. Explain why V(4) = 0 even though 4 is a positive number.

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Exit Ticket

A sheet is 12 in. by 12 in. Squares of side x are cut from the corners and the sides folded up.

Volume function: __________________________ Contextual domain: ________________________

Explain why the upper limit is 6: ____